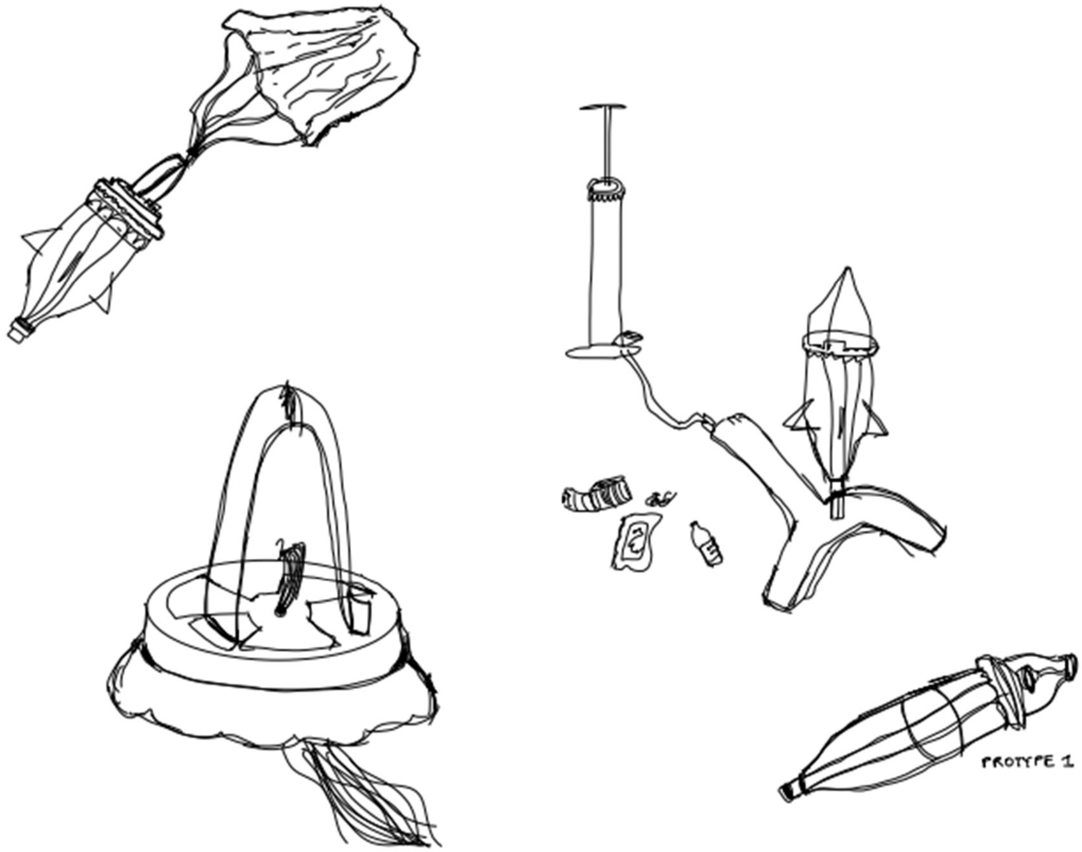

Department of Mechanical and Mechatronics Engineering

ME100: Water Rocket Toy Design Project

A Report Prepared For:
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Salutation,

ME100 is the quintessential Mechanical Engineering course. A preparatory experience that gives all of its students a breadth of what later years and onsite experience has to offer. While our group of four freshman Mechanical Engineering students can come to agreement that some of the assignments can get rather tedious, the culmination of everything that we have learned is very valuable. The following report entitled ME100: Water Rocket Toy Design Project, is no different than the description above. In whole, the report gives an in-depth understanding of the design process for our toy, explaining its key aspects and the trials and tribulations that were thought through along the way.

The educational elements of ME100: Introduction to Mechanical Engineering Practice I, are closely displayed through the work of our toy design project. For instance, this whole report effectively demonstrates engineering communication, a key component of the DCAP side of our course. The CAD models, attempt to construct sound structures and models show key aspects of the EGAD side of this course. While our attempts at displaying our knowledge are mediocre at best, these are only our first steps and, like a baby, we will be able to run eventually. The next 15 or so pages will give a detailed analysis of the viability of our toy, but we believe it is more than imperative to note that working on this project, successful or not has been a fun learning experience and made us feel like rocket scientists, even though we were far from being one.

This work was completed entirely by the undersigned, and has not been submitted for credit at this or any other institution. Thank you for taking the time to review this work. If you have any questions or concerns, please do not hesitate to contact either of us

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Best Regards

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Summary

This report follows the design process of creating an automatic parachute deployment system for a typical bottle rocket targeted towards kids age 8-12, ultimately deciding whether or not this product can be viable as a toy. Although many bottle rocket models currently exist, few opt for mechanically timed parachute deployment mechanisms, and instead use electronics to do so. The advantage of a mechanically timed launch system is that it is more child-friendly, as it does not require any pre-requisite knowledge to understand how it functions.

When designing this mechanism, three different potential solutions were considered; a pressure-based mechanism, a mechanism that worked with gravity to unload the parachute and finally a mechanically timed deployment. Although the pressure-based and gravity-based solutions seemed viable, after some research it was concluded that the mechanical timer was the only viable option to employ in the design.

Upon deciding how the toy works, the main functions of the toy were decided. First, the toy nose cone that houses the timer must fit onto any typical two-liter plastic bottle and the cone also must be able to house all the components that launch the parachute. This includes the timer itself, a parachute, and spring that ejects the parachute outwards. Within the nose cone exists two more functions related to the timer, a wind-up sequence which starts the timer and an initial trigger phase which starts the timer at the launch of the rocket.

After considering all the necessary functions, the final product to date is constructed from a 3D printed nose cone that fits on any bottle. Within the nose cone is the parachute, spring, and timer. The timer utilized is called a tomy timer which can be found in any typical wind-up toy. The spring used to launch the parachute is a bendable strip of plastic that springs outwards once the nose cone falls off. Upon trial testing, it was determined that both the timer, trigger and spring mechanisms were viable as the rocket launched, and deployed the parachute during its descent. Although this is a working solution, there are still a few more remaining challenges, the mounting of the tomy timer must be more secure as well as the plastic spring.

1.0 Introduction

This report follows the development of a bottle rocket that employs a mechanical timer to deploy a parachute upon the rocket's descent. Since bottle rockets are a common hobby for many, there are many existing and viable designs for the launch system of the rocket and the design of the rocket itself. As a result of this, this report will solely focus on creating the deployment mechanism for parachutes and capsule that contains it because only few existing designs use mechanical deployment mechanisms. Some terminology that will be referenced throughout the report are: apogee (max height of rocket), (seen in figure 1) nosecone top and base.

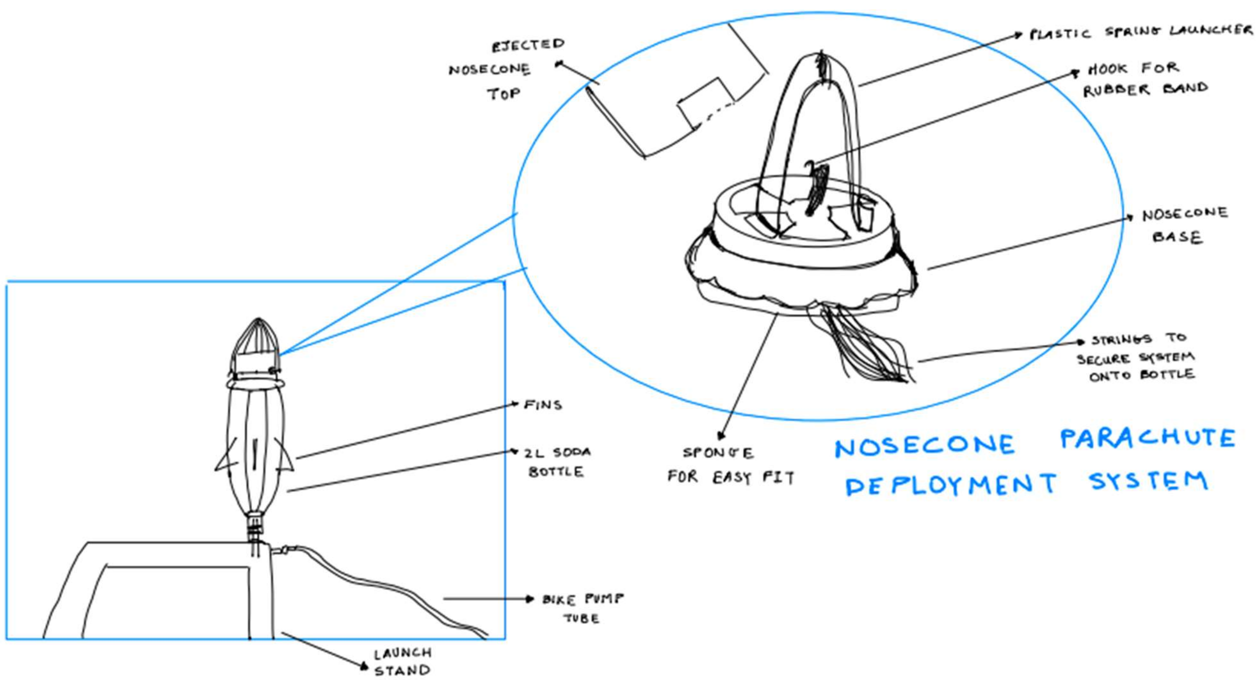

Figure 1: Rough diagram detailing the key components to the Nosecone Parachute Deployment System.

1.1 Play Pattern / Age Range

The target audience for this toy are children in the age group of 8-12. This is because it will be a part of a kit which the consumer will put together. Since putting it together requires slightly higher cognitive ability, the age range for the toy will be higher as a result. The play pattern of the toy follows a onetime construction manual which instructs the user on how to build a reusable bottle rocket. Following the building phase, the rocket can then be launched outside for enjoyment.

1.2 Comparable Products

Bottle rockets have existed for centuries, with the first being built in around the 12th century AD. As a result, commercial rockets and hobby rockets have existed for many years, and therefore there are many variations of designs that exist. Currently there are many different models that are available on the market, ranging from more complex bottle rockets for hobbyists to simple models for children. However, the toy being designed needs to be more aligned with the child variations – not only because children are the target audience, but also because the nosecone needs to be simple enough that it can be attached to any product.

Below in figure 2 is a bottle rocket that fits into the description above of similar products. It is an educational kit that is clearly aimed at children. Furthermore, it appears to also use a generic 2L bottle and is sold in the form of a kit which mirrors the design of the rocket detailed in this report.

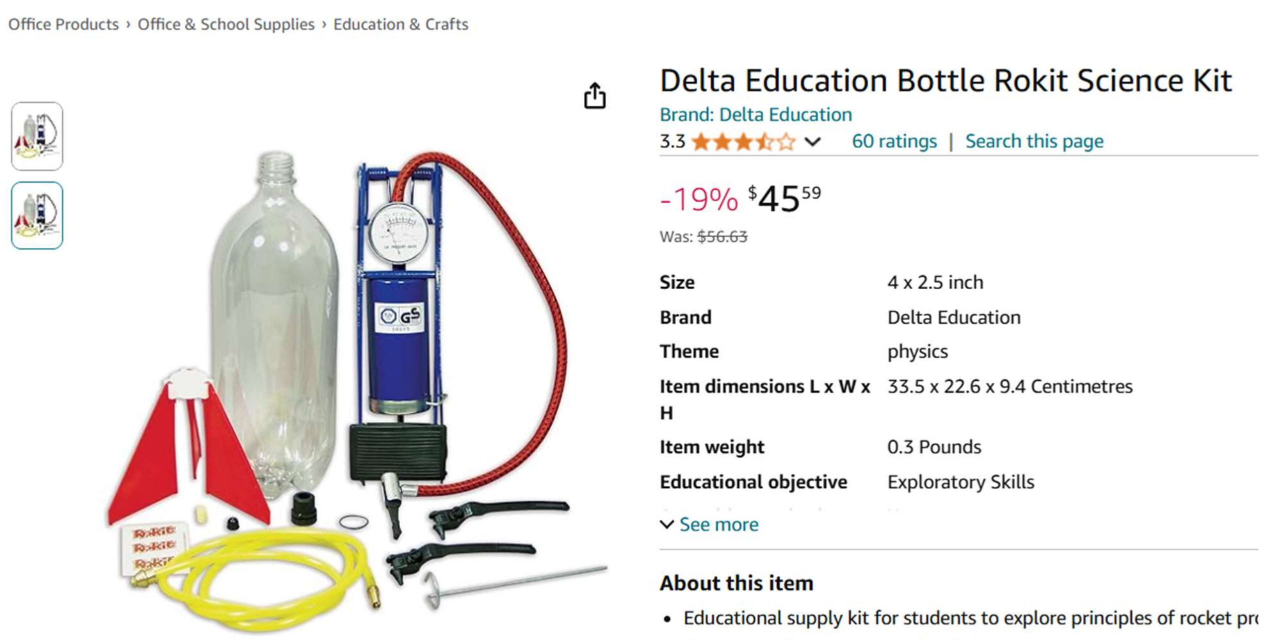

Figure 2: Similar product to the water rocket detailed in this report

1.3 Objective

The objective of this report is to detail the complexities of a nosecone attachment for a 2L water rocket with a built-in parachute deployment system and whether or not this product is suitable as a toy that can perform multiple launches reliably.

2.0 Problem Definition

Water rocket parachute deployment mechanisms are commonly discussed through many forums as they can often be overcomplicated or unreliable. For this reason, parachute deployment is the problem being solved in this report through the development of a modular nosecone attachment that reliably deploys a parachute at apogee and is adaptable to any standard 2L bottle.

2.1 Required Functions

Table 1: Key functions of the parachute deployment nosecone.

Function Component	Fit of nosecone base onto 2L bottle. Nosecone top fit onto nosecone base	Mechanical timer releasing nosecone top at specific point.	Trigger that starts mechanical timer	Spring to knock nosecone top off base and deploy parachute
Explanations	The nosecone base accepts any bottle in the 2L range. The nosecone top fits securely on the nosecone base.	Mechanical timer that has a windup sequence which delays parachute deployment.	A trigger that activates the timer as the rocket is launched.	A spring that propels the nosecone top off the nosecone base once the timer expires and deploys parachute

2.2 Constraints and Criteria

Table 2: Key constraints and criteria of the parachute deployment nosecone pertaining to its functions.

Constraints	Nosecone must be able to be swapped from bottle to bottle at least 5 times.	The timer must expire in under 3 seconds.	The timer trigger must start the timer in under 0.2 seconds of the launch.	Must have at least a 95% success rate of deploying parachute and nosecone.
Criteria	Optimizations can be made in term of the fit's reliability. The nosecone should be as secure as possible before falling off.	The timer should minimize the period between the apogee being reached and it going off as much as possible	The time between the timer triggering and the launch happening should be minimized.	Spring should maximize the distance at which it knocks the nosecone off and parachute away.

Explanations/ Justification for constraints and criteria	The nosecone being able to be swapped at least 5 times is reasonable as one bottle lasts around 30 launches so a 150 launch lifespan is reasonable.	Since the time to apogee is under 3 seconds, the timer should be fast and unwind under 3 seconds.	0.2 seconds is a necessary constraint for the trigger mechanism, as the time to apogee is just around 2.5 seconds and the timer takes around 1.5-2.5s to deploy the parachute.	A parachute deployment nosecone without a reliable spring is no good. Therefore >95% success rate of deployment is something that should be upheld.

3.0 Technical Progress

3.1 Solutions Considered

The project involves designing a parachute deployment device for a water-pressure-powered bottle rocket. The initial concept is centered on housing the deployment mechanism within a two-piece nosecone consisting of a base and a detachable cone. The original design used a friction fit to secure the base of the nosecone onto the bottle, but this proved unreliable when adapting to differently shaped two-liter bottles. To address this issue, a more flexible solution pictured in Figure 3, using a sponge and strings to secure the nose cone was implemented, ensuring compatibility and stability across various bottle shapes. However, a component of this design was noted to be faulty after various tests were performed on the bottle rocket. The strings that would hold the nosecone base onto the bottle would constantly come lose and cause the nosecone to wobble around. The initial solution of using a zip tie as a way of securing the strings pictured in figure 3 was then improved to a 3-D printed clip that would hold the strings and nosecone base into place using a simple nut and bolt tightening method (see figure 4). This addition proved much better than rubber bands as it does not loosen as easily, is simple to install, and withstands various testing. With the nose cone design finalized, the focus shifted to determining the most effective deployment mechanism.

Figure 3: Sponge solution that provides smug fit between nosecone compartment and 2L bottle. Also pictured is the old iteration of securing the strings to the nosecone base using rubber bands.

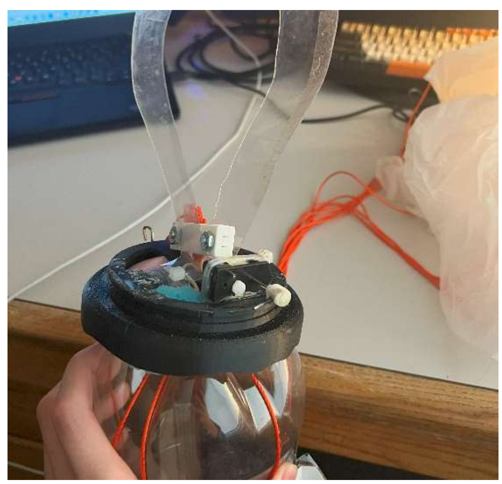

Figure 4: Pictured is the new clip solution that is used to secure the strings to the nosecone base.

The proposed solution for the deployment of the parachute is a straightforward spring-loaded mechanism (see figure 5) where once “deployed”, the spring would expand and push off the nosecone while pushing out the parachute. First, metal springs were considered for the design but they are not simple to work with as they cannot be compressed easily and none were found to be long enough to launch the cone to a safe distance from the rocket. A thin strip of plastic ended up being the ideal material for the spring, specifically PETG [1] due to its high durability and flexibility. Notably, it could be compressed easily into a small ball, something the spring could not do, and contain the perfect amount of force to knock the cone off of the base while deploying the parachute.

Figure 5: Final nosecone solution, depicted is the nosecone base, plastic spring, tomy timer and attachment string

The main criteria is for the parachute to deploy as near to the apogee point of the rocket’s flight. Three potential solutions were explored for this purpose (see figure 6): a gravity-based system, a pressure plate design, and a timer-based mechanism. The gravity-based mechanism would use some sort of weight which would react to change in bottle orientation or direction of movement. Unfortunately, this system would not work as it would only be able to detect when the bottle stops accelerating upwards [2]. The pressure plate mechanism would utilize a pressure plate which upon launch, would be pressed down by the air pressure exerted onto the nose cone when the bottle is in upward motion. This plate would be released at the apogee where the velocity is equal to zero, and the parachute deploys. Both the gravity-based system and the pressure plate design were too susceptible to premature deployment due to their reliance on consistent and predictable flight orientation.

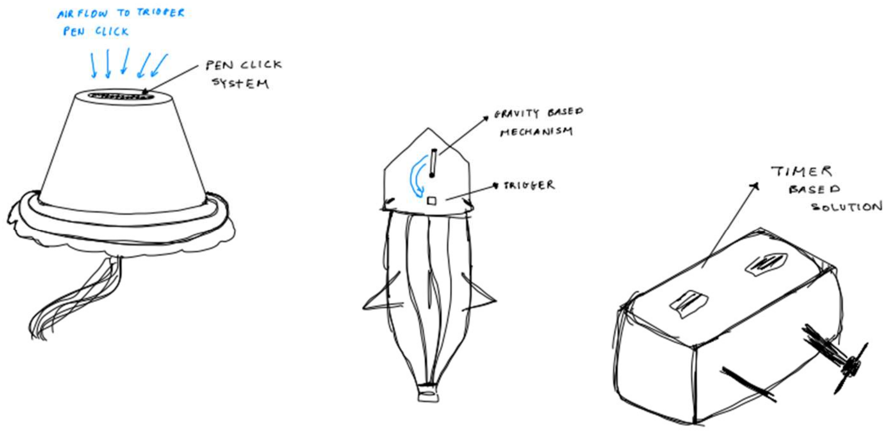

Figure 6: Launch timer concept sketches. From left to right: pen click, gravity-based, and tomy timer.

The timer-based mechanism, as seen in Figure 6, would use a simple wind-up timer from a wind-up toy and be attached to a rubber band which would be securing the nosecone top to the nosecone base. When the timer finishes unwinding, the rubber band would be released and the spring would knock the nosecone top off and deploy the parachute. Ultimately, the timer-based solution was selected for its superior reliability as it did not rely on flight orientation, and due to the fact that it could be set to match the time to apogee, releasing the nosecone as close to this point as possible, a key constraint to this function.

For the timer-based solution to work, the timer needs to be started the moment the rocket launches. Two solutions were devised, one being a pin that holds the timer's crank arm in place and falls out during launch or another where a physical stop is built onto the launcher so that when the rocket launches, its no longer engaged and the timer starts. The two designs were combined (see figure 7) into a single design where there is a block that is inserted as a physical stop, like the pin, and this block is tied to the launcher so it would get pulled out during launch. This decision was made because the pin idea could not produce a false start even if the rocket shifts during the prelaunch phase as the pin would be secured to the actual rocket. The rocket stand idea was implemented as the moment the rocket flies into the air, the disengaging of the pin would happen automatically as the string that holds the pin in place is tied to the bottom of the launcher.

Figure 7: Operational setup, depicted is the timer trigger mechanism, full nosecone ready to launch, with tomy timer queued for launch.

3.2 Testing results

Testing of any rocket is unpredictable, for this reason, there was emphasis on calibrating the tomy timer before sending the rocket out on the field for full scale testing. The first step to calibrating the tomy timer was done using a slow-motion camera. In order to understand how the rubber band that releases the nosecone top interacts with the tomy timer, a series of snapshots were taken of the timer as it wound down (see figure 8). These results were crucial, as they gave an understanding of when the timer released the nosecone top which would then be used in later calibration testing seen below.

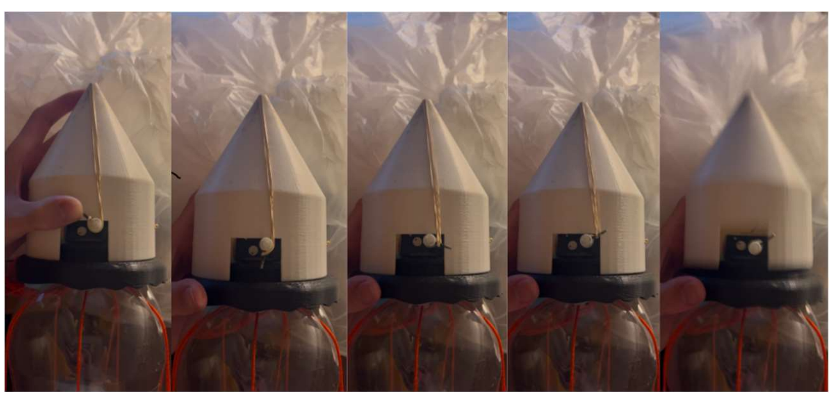

Figure 8: Visual representation of the test done to see where the rubber band releases the nosecone as the timer unwinds.

Analyzing the photo images, shows that the rubber band is released near the 2 o'clock point which is expected of this interaction.

Using the information gathered above, then allows for a precise understanding of how long the timer delays the parachute deployment during the launch cycle. By winding the timer to predetermined queue points (see figure 9) that were a certain amount of rotation from the release point. Then recording the time is takes for the timer to complete that amount of rotations using a stopwatch, the precise time it takes for the timer to release the nosecone during launch was calculated (see table 3).

Table 3: Tomy timer calibration data, turns queued vs. time to end

Turns Queued	Time Taken (s)
0	0
0.5	1.60
1	2.65
1.5	3.89
2	5.30
2.5	6.87

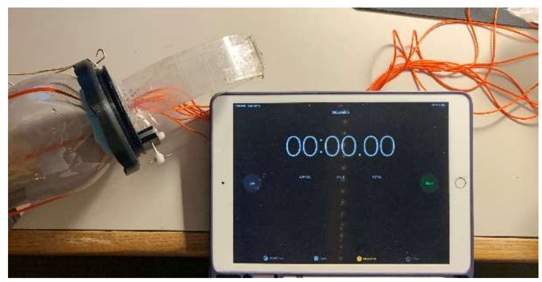

Figure 9: Tomy timer calibration setup

The data recorded above was then implemented during field testing. By first launching the rocket without the nosecone attachment and recording the data seen in table 4 trial 1 and 2, the approximate time to apogee was determined. Taking this into account, the timer was then queued 1 rotation as predetermined to match the time to apogee by the tomy timer calibration results. By filling the bottle with 40 psi and 500 mL of water during all trials, the testing, seen in figure 10, proved that queuing the timer 1 rotation would fulfill the criteria of having the timer go off as close to the apogee point as possible. This testing serves another purpose specifically in terms of the distribution of the toy. By determining the optimal queue amount for the best launch, users could be advised of this amount to promote the best experience during playtime.

Table 4: Flight trials with respective time taken for rocket to reach apogee.
Trial 1 and 2 are without the nosecone attachment and 3,4,5,6 are with the nosecone.

Flight Trial #	Time to Apogee (s)
1	1.83
2	2.03
3	2.27
4	2.30
5	2.48
6	2.31

Figure 10: On field testing of water rocket, video analysis used to determine time to apogee.

3.3 Progress to Date

Despite the various changes in solutions shown through the multiple prototypes, all important components have been integrated into the current product. Seen in figure 7 above, all the nosecone, base, and attachment fixtures have been implemented and tested through numerous launch trials, with the string/sponge solutions working very well to give the rocket hot swappable capabilities on top of its secure fit. The tomy timer solution has been reliable for the test runs and has released the nosecone at optimal times during the launch with little error. As for the parachute deployment system, the general idea has held together well during testing with the choice of a plastic spring opposed to a metal spring being as good as excepted since the force delivery is optimal and consistent. In terms of the calculational work, the timer has been calibrated to find the optimal wind-up amount and the time to apogee has also been measured to match it with the timer calculations as much as possible. It is safe to say that this water rocket design process has gone well as all key components have been implemented and tested.

3.4 Remaining Challenges

Most key functions were proved viability shown through testing and implementation above, however, fine details were left unfinished with some quick solutions being used in their place. For instance, a challenge that was encountered many times during the design implementation phase was the mounting of the spring and tomy timer. A proper solution was not utilized before the works-like product was completed, rather, glue and zip ties were used in their place. While the product worked under this solution, the mounting was very inconsistent and the glue would often crack under load. To solve this issue, the nosecone base should be modified to contain a sort of bracket (see figure 11) to contain both the plastic spring and tomy timer. The concept sketch below shows roughly how the clip and friction fit slot would work as a mounting mechanism. This would make the toy much easier to assemble and make the parts hot swappable for replacements.

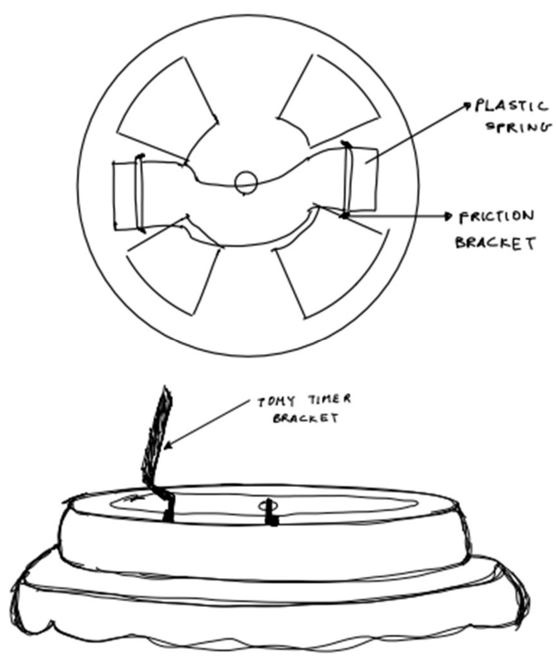

Figure 11: Concept sketch of tomy timer bracket and the plastic spring friction fit slot.

Another challenge that was encountered during the testing phase was the reliability of the parachute. While the parachute would deploy on each test, the rate at which it would catch air and expand varied heavily. After all tests were concluded and video recordings were analyzed, it was determined that the likely reason for these inconsistencies was the type of material used. For this reason, the making of a new parachute with a material that sticks to itself less is something that still needs to be refined.

4.0 Conclusion

4.1 Conclusions

Implementing a modular nosecone attachment to a 2L bottle rocket with an internal parachute deployment system is a viable option as a toy, meeting a crucial objective set for this project. In general, safety was a priority when designing the toy and it exceeded expectations as the rocket could be pressurized and launched from a distance. Additionally, the main function of the nosecone, deployment of the parachute, prevented the rocket from falling rapidly, providing an extra layer of safety. As a summary, this rocket consists of a 3D printed nosecone attachment, featuring a detachable top and base, which consistently deployed the parachute at or near apogee. The tomy timer solution offered reliability by avoiding dependence on flight path, orientation or external forces. Additionally, the PETG plastic strip used as a spring is one of the viable options for the deployment system, as it is both durable and flexible. A significant portion of the testing time was spent in calibrating the tomy timer to set off and release the nosecone at or near apogee. With its successes, proven by the test runs, certain problems did arise. For instance, key parts of the rocket kit such as the spring and tomy timer which had been mounted by glue, cracked under stress. Recommendations of a bracket that holds these components in place would drastically increase the durability of the rocket and make it easily hot-swappable.

4.2 Recommendation

Few recommendations for the project can be made in terms of refining the design process of a few components. Providing a few plastic springs in the kit as replacements is certainly something that can be considered in the final product of the toy. During the stress testing and trials performed on the rocket, it was noted that the plastic spring lost some of its elasticity over time. This does not hinder with the constraints mentioned earlier as the spring should only withstand 20 trials. However, to promote the constant reusability of the toy, additional springs should be provided. Additionally, a failsafe for the parachute not deploying is something that should be considered for a more refined design. The launching of a water rocket is inherently dangerous but implementing a mechanism that guides the rocket down in case of a parachute deployment error would certainly make it more kid-friendly. Nonetheless, this water rocket journey proved successful and is deservingly called a toy. It is something that was very fun to make and any kid would certainly enjoy to have. The combination of building a rocket, and being able to see it perform a launch is both exhilarating and satisfying. This product would certainly be a hit at any store that it is sold at.

References

[1] A. Plastics, “What is PETG?,” ACME Plastics, Inc., https://www.acmeplastics.com/what-is-petg?srsltid=AfmBOopvq17fAqFaDGhpLnonWTCVAC1oDf1krIDBGmaYphAO6Okk4kQK (accessed Nov. 12, 2024).

[2] “gravity based system for water rockets,” YouTube, https://youtu.be/oC8AbWXFiTw?si=Ye2OGy-74vt5xDxD (accessed Nov. 17, 2024).